



B3. Planning under Uncertainty

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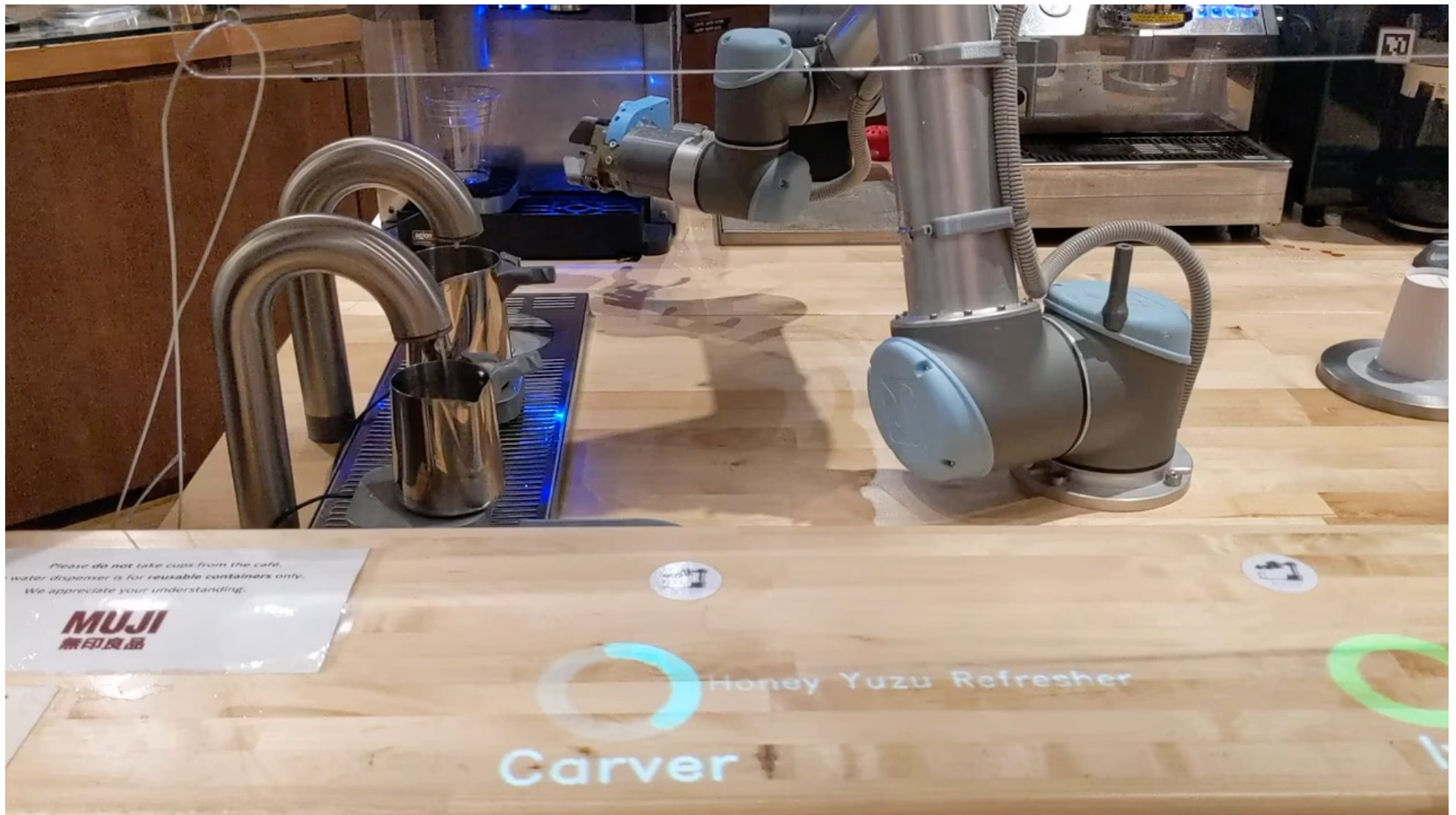
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Computing



By now, we know...

- Hierarchy in Robot Planning
 - Symbolic Planning
 - Task – Motion Planning
 - Motion Planning
 - Assumption: Known deterministic world
 - This lecture: Relax the perfect model assumption
 - Aim: Robust & resilient robot operation
-



<https://www.youtube.com/watch?v=DHqLkGPrzso>

Planning under Uncertainty

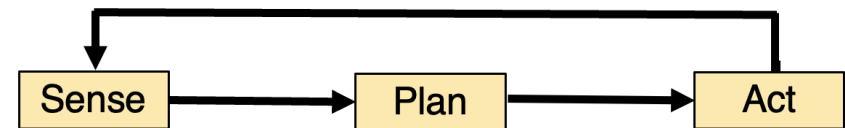
Goal: Generate strategies for the robot to achieve its objective (e.g., reach a goal, serve Yuzu Honey Refresher, gather good data, etc.) **well**, despite various types of uncertainty

Well: Can be many criteria

Planning under Uncertainty

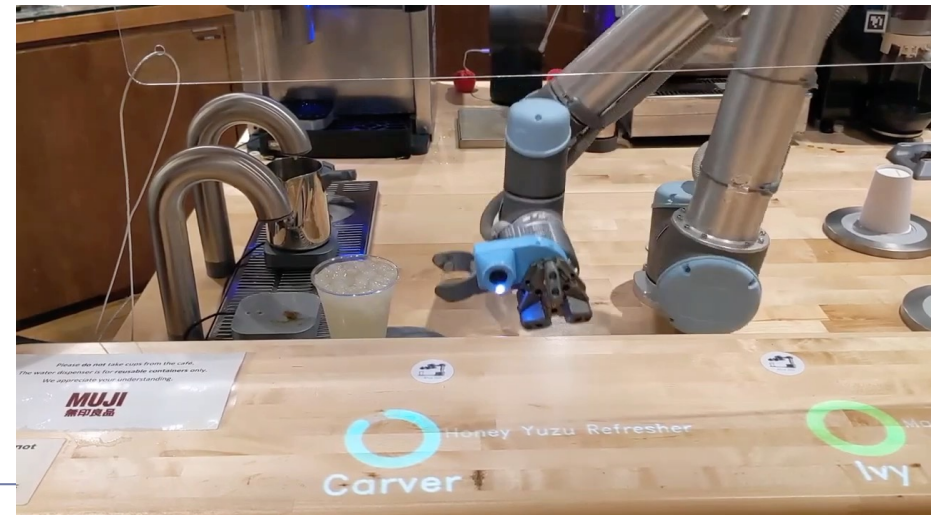
Sequential decision-making ;
Have a model to predict
outcomes of actions/sensing
before execution

Classify: The act and the sense



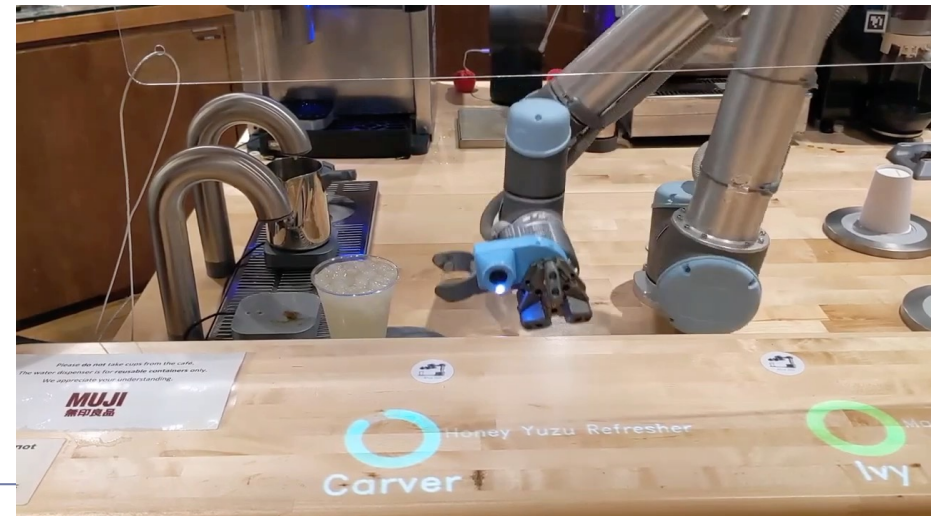
A bit of terminology

- Uncertainty in the act
 - Deterministic vs non-deterministic effects of actions
 - Does an action always move the robot from a state to exactly one (other/same) state?
 - Yes: Deterministic
 - No: Non-deterministic
- The video example?



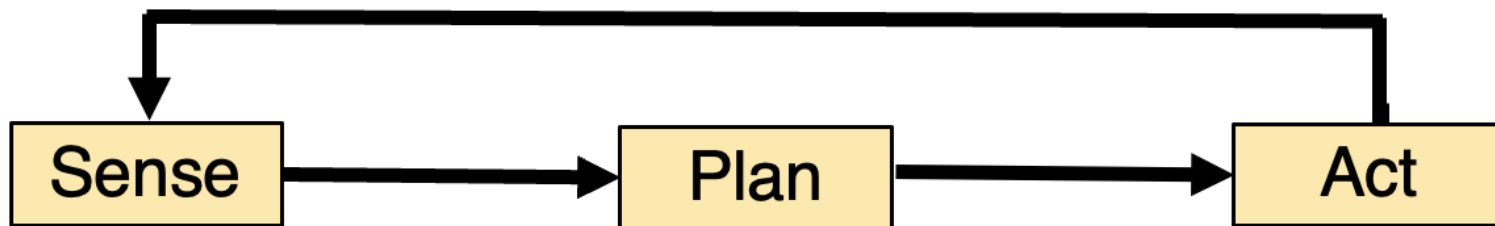
A bit of terminology

- Uncertainty in the sense
 - Fully observed vs partially-observed
 - Does the robot know the state of the world exactly?
 - Yes: Fully observed
 - No: Partially observed
- The video example?



Seems only 2 types of uncertainty, but...

- Covers all interactions between the planner and the system outside of the planner, incl. the environment



- Each of the Sense and Act can also be expanded
 - For instance, when human is the one that will execute the plan, then the outcome of Act can include errors about which action is executed

Seems only 2 types of uncertainty, but...

- Can represent various types of uncertainty, e.g.:
 - Actuators errors
 - Measurement errors
 - Errors in processing sensor data
 - Errors in modelling the agents' own dynamics and the environment
 - Uncertainty in human behaviour when a robot collaborates with humans
 - Initially unknown physical parameters (e.g., coef. friction, weight, etc.)
 - Trick is in problem modelling
-

This session

- Frameworks for planning under uncertainty
 - Markov Decision Processes (MDPs)
 - Partially Observable MDPs (POMDPs)
 - Reinforcement Learning
 - Offline Solving: Value Iteration
 - Value Iteration for MDP
 - Online Solving: MCTS-based
 - MDP
-

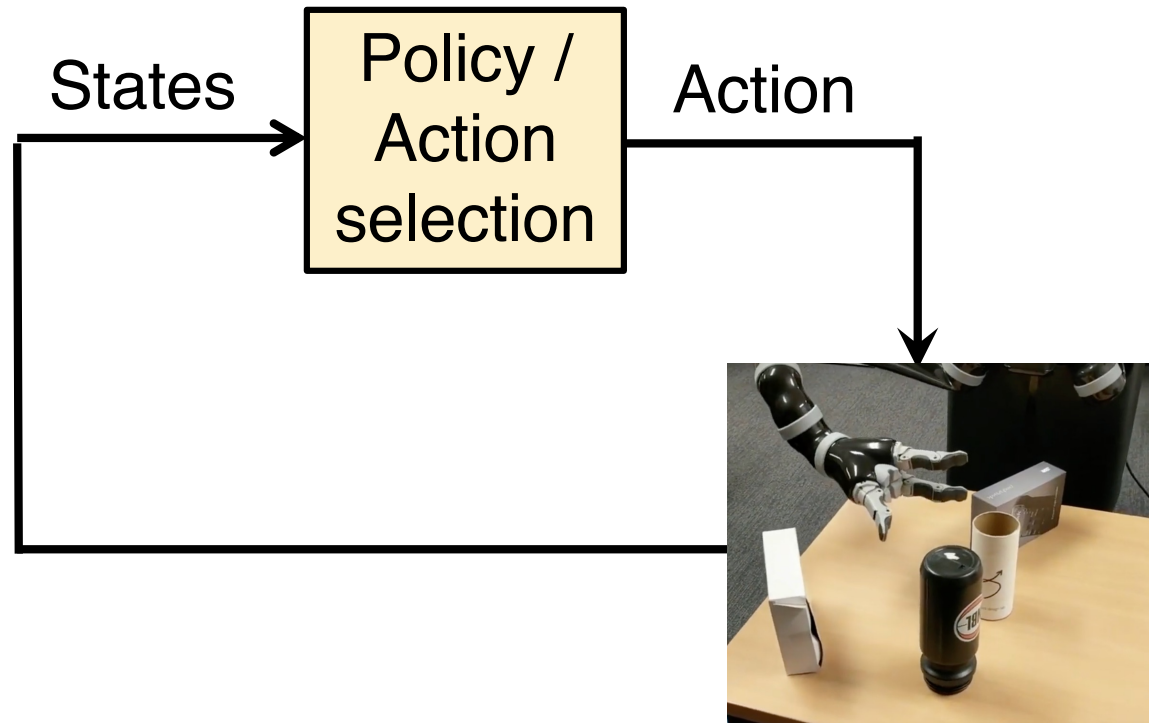
Frameworks for planning under uncertainty

- Markov Decision Processes (MDPs): When the effects of actions are non-deterministic
 - Partially Observable MDPs (POMDPs): Uncertainty the effects of actions are non-deterministic and the world is partially observed
 - Reinforcement Learning: When the models are not initially available
-

Markov Decision Processes (MDPs)

- Decisions are made at discrete time
 - At any given time step, the agent's state is known exactly (fully observed)
 - But, the effects of actions are not known exactly before the action is executed (non-deterministic action effects)
 - How does this affect the format of the planning solution?
Would a path / sequence of actions be a sufficient plan?
 - No, we need to compute a mapping from states to actions, namely a policy
-

Markov Decision Processes (MDPs)



$T(s, a, s')$: Transition function, cond. prob. func $P(s' | s, a)$
where s, s' : States, a : action

R : Reward function

Defining an MDP Problem / Model

- Formally defined as a 4-tuples

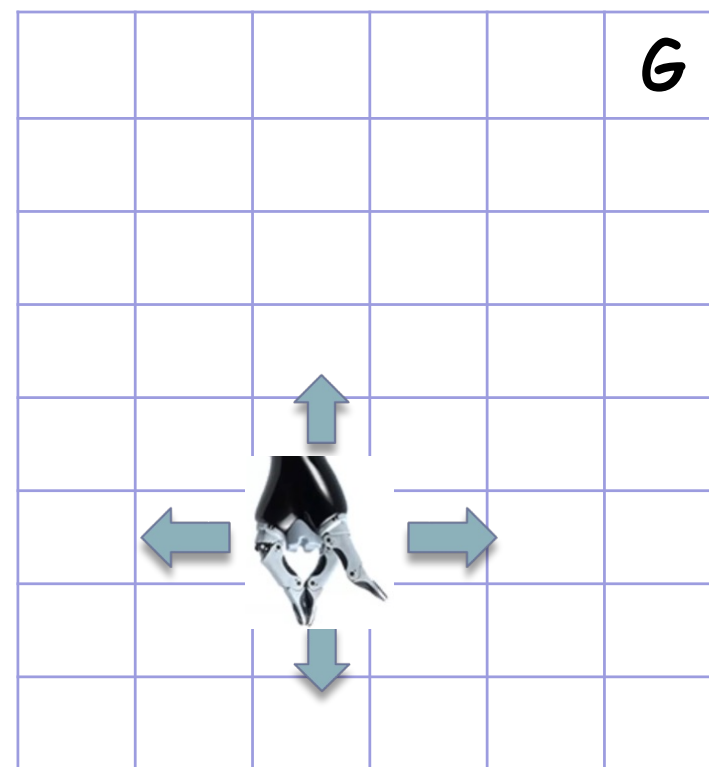
(S, A, T, R):

- S: State space.
- A: Action space.
- T: transition function.

$$T(s, a, s') = P(S_{t+1} = s' \mid S_t = s, A_t = a).$$

- R: Reward function.

$R(s)$ or $R(s, a)$ or $R(s, a, s')$



The states & actions can be continuous spaces
In this case, the transition is the pdf

Markov Decision Processes (MDPs)

- Solving an MDP = finding the best policy π^*
- What does “best” means?
- Many objectives have been proposed
- Some commonly used are the policy π that maximizes

- Myopic: $E[R_t(s, a) | \pi, s]$

- Finite horizon: $E \left[\sum_{t=0}^k R_t(s, a) \mid \pi, s_0 \right]$

- Infinite horizon: $E \left[\sum_{t=0}^{\infty} \gamma^t R_t(s, a) \mid \pi, s_0 \right]$

s_0 : initial state

s : state

a : action

$R_t(s, a)$: immediate
reward

γ : discount factor with
 $0 < \gamma < 1$

Difference in Optimal Policy?

- Finite vs Infinite Horizon?

$$E \left[\sum_{t=0}^k R_t(s, a) \mid \pi, s_0 \right] \quad \text{VS} \quad E \left[\sum_{t=0}^{\text{inf}} \gamma^t R_t(s, a) \mid \pi, s_0 \right]$$



s_0 : initial state

s : state

a : action

$R_t(s, a)$: immediate
reward

γ : discount factor with
 $0 < \gamma < 1$

The optimal policy is stationary

Value of a policy

- The expected total reward the agent will gather if it executes a policy π is called the value function and is denoted by V_π .

$$V_\pi(s) = \boxed{R(s, a)} + \gamma \boxed{\sum_{s'} T(s, \pi(s), s') V_\pi(s')}$$

Immediate reward

Expected total future reward

Optimal Value function

- Each policy has a corresponding value function.
- An optimal policy π^* is a policy whose corresponding value function is the maximum.
 - Meaning at each state, $V_{\pi^*}(s)$ is higher than (or at least the same as) V_{π} for any other policy π .
 - V_{π^*} is often simplified as V^*
 - In other words,

$$V^*(s) = \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V^*(s') \right)$$

$Q(s, a)$



A note

- There is a unique optimal value function V^*
- But, there might be more than one optimal policy
 - If we know V^* , the optimal policy can be generated easily

$$\pi^*(s) = \operatorname{argmax}_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V^*(s') \right)$$

Markov Decision Processes (MDPs)

- Decisions are made at discrete time
 - At any given time step, the agent's state is known exactly (fully observed)
 - But, the effects of actions are not known exactly before the action is executed (non-deterministic action effects)
 - The solution format is a policy –that is, a mapping from states to actions
-

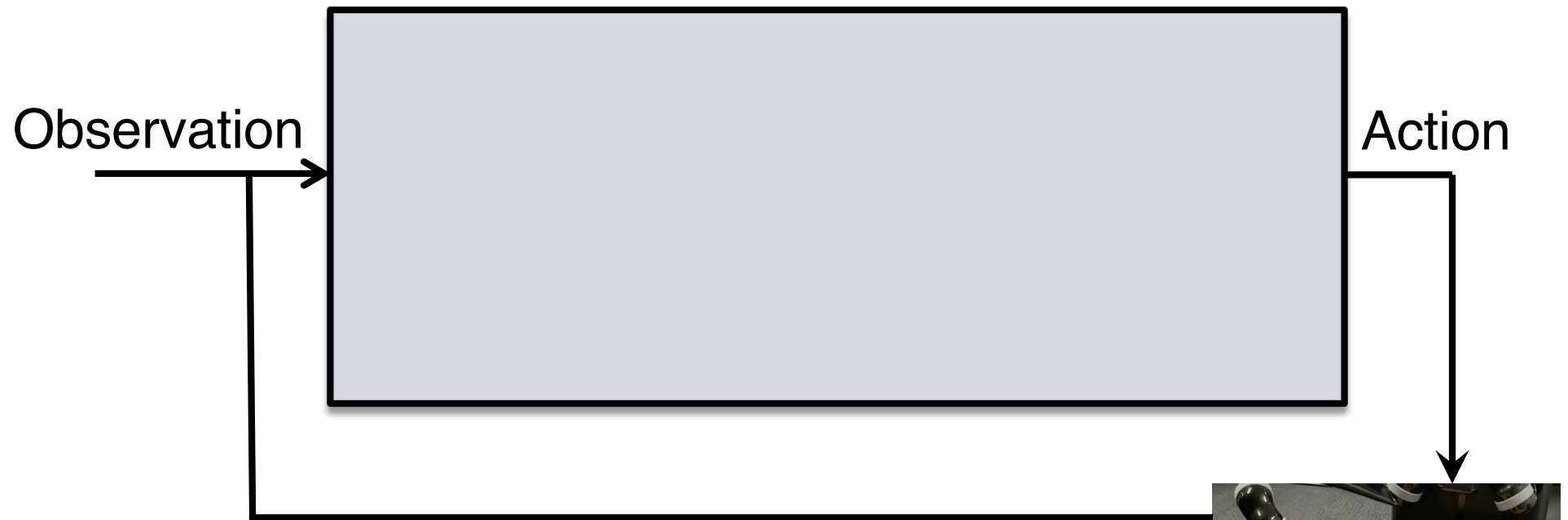
This session

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 - ✓ Markov Decision Processes (MDPs)
 - Partially Observable MDPs (POMDPs)
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Partially Observable Markov Decision Processes (POMDPs)

- Discrete time step
 - The effects of actions are not known exactly before the action is executed (non-deterministic action effects)
 - In addition, at any given time, the agent's state is NOT known exactly (partially observed)
-

Partially Observable Markov Decision Processes (POMDPs)

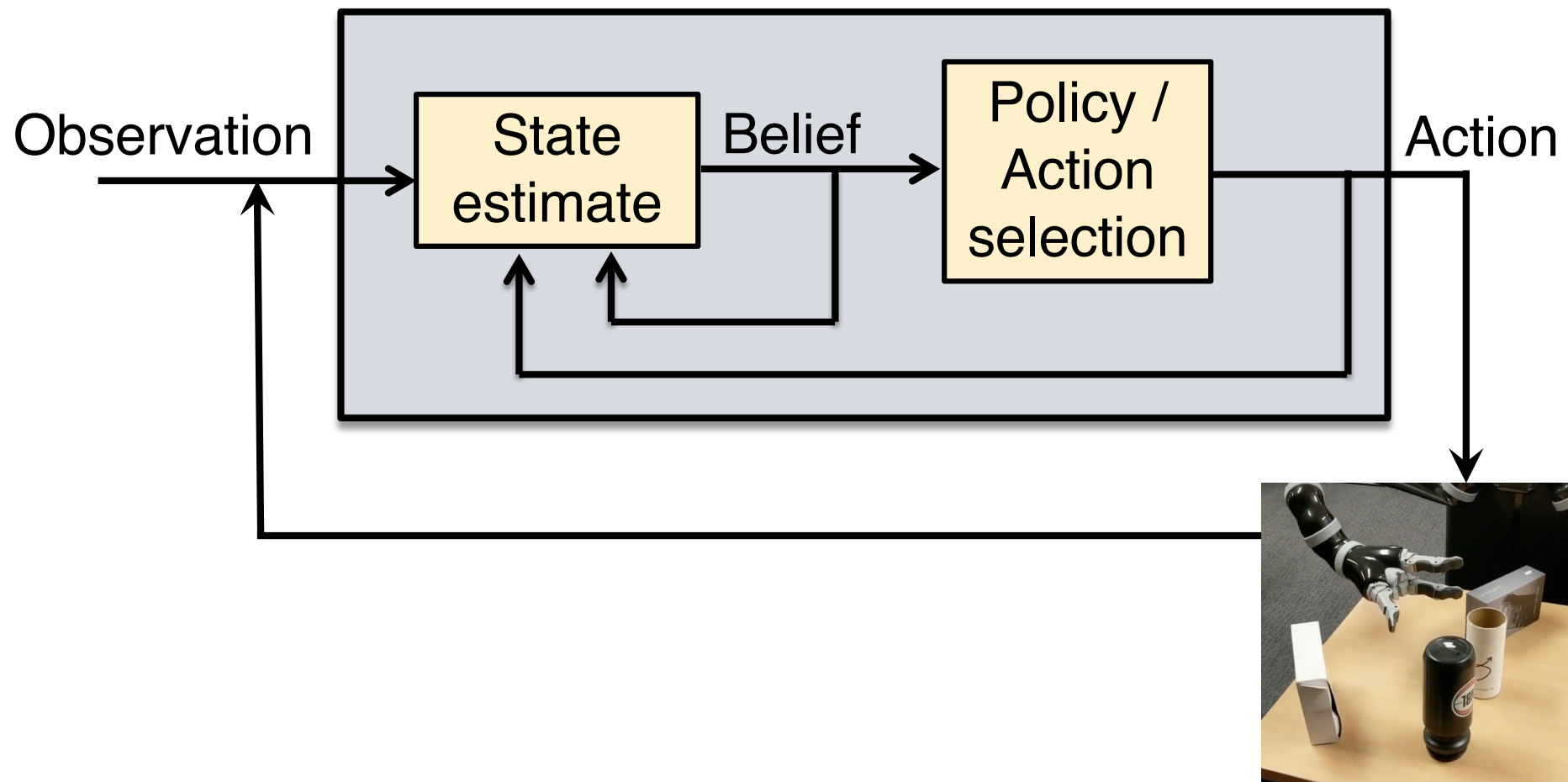


$Z(s', a, o)$: Observation function $P(o \mid s', a)$; $o \in O$
 s' : States, a : action, o : Observation

$T(s, a, s')$: Transition function $P(s' \mid s, a)$
 s, s' : States, a : action

R : Reward function

Partially Observable Markov Decision Processes (POMDPs)

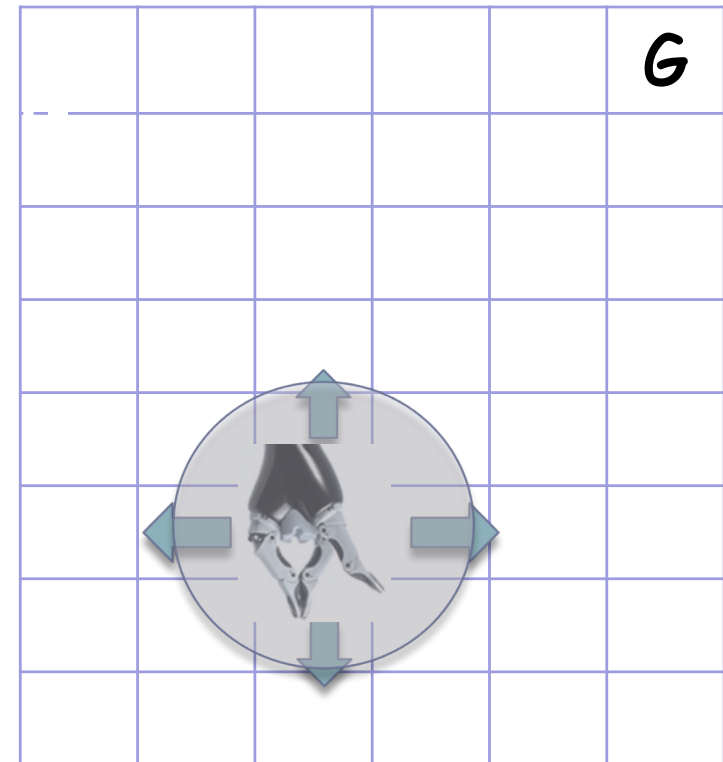


Belief: Distribution over states

Solving a POMDP: Computing the best policy –maps beliefs to the best action

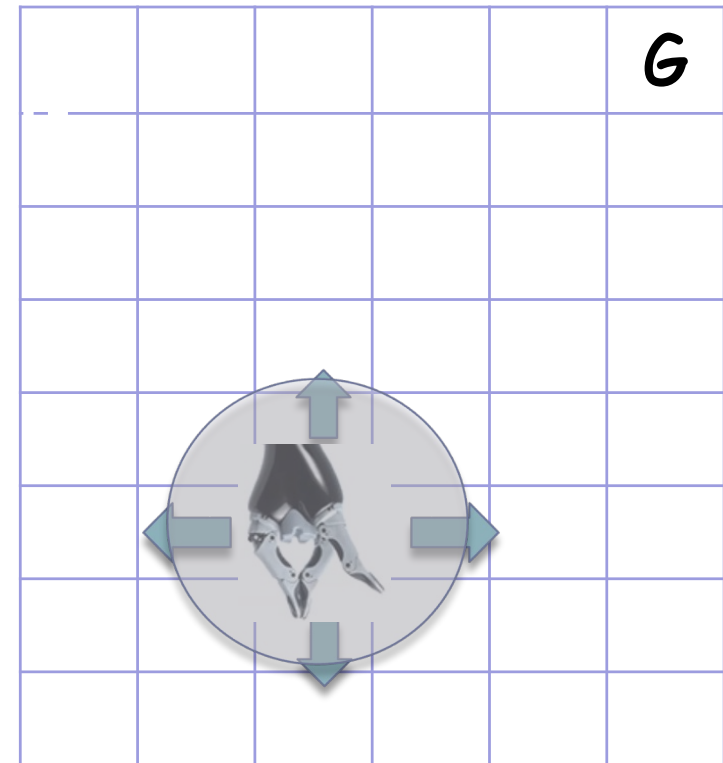
POMDP Model

- Main components:
 - State space (S)
 - Action space (A)
 - Observation space (O)



POMDP Model

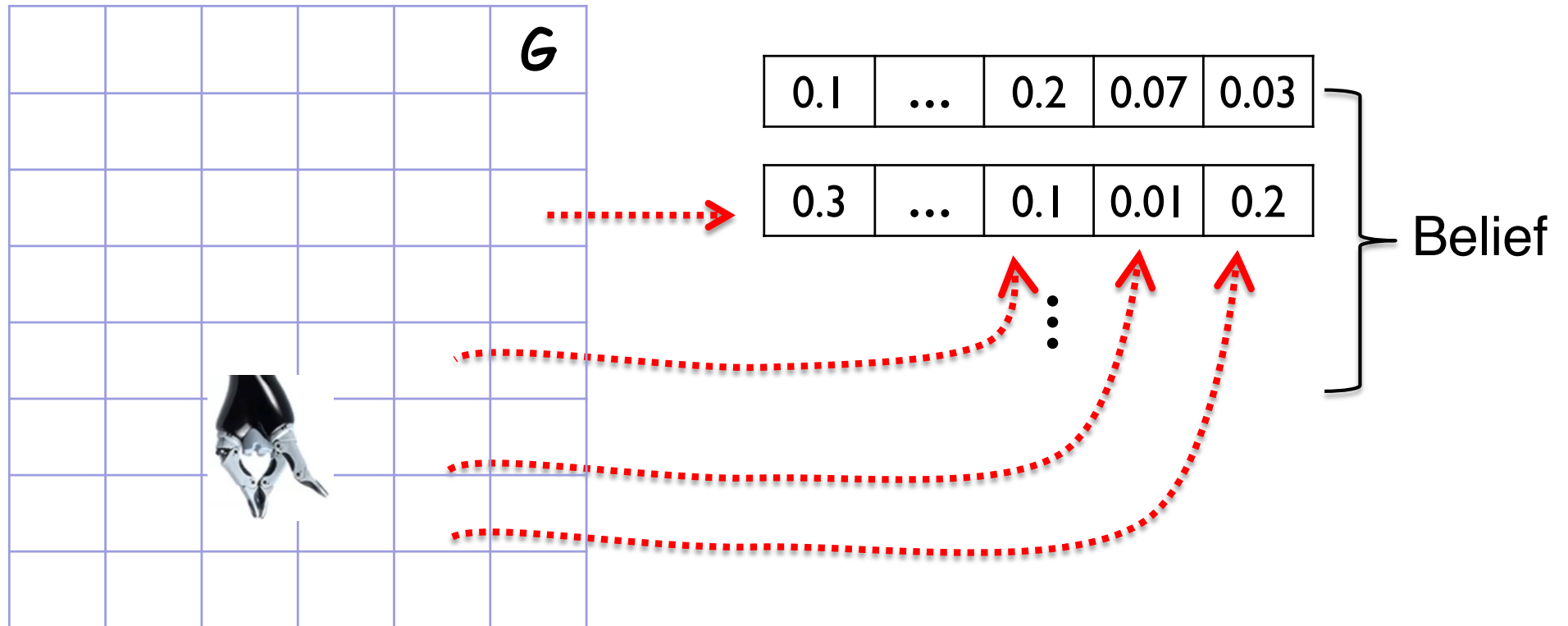
- Main components:
 - State space (S) ← Not known
 - Action space (A)
 - Observation space (O)
 - Transition function (T)
 - $T = P(s' | s, a)$
 - Observation function (Z)
 - $Z = P(o | s', a)$
 - Reward function (R)
 - $R(s, a)$



The states, actions, & observations can be continuous spaces

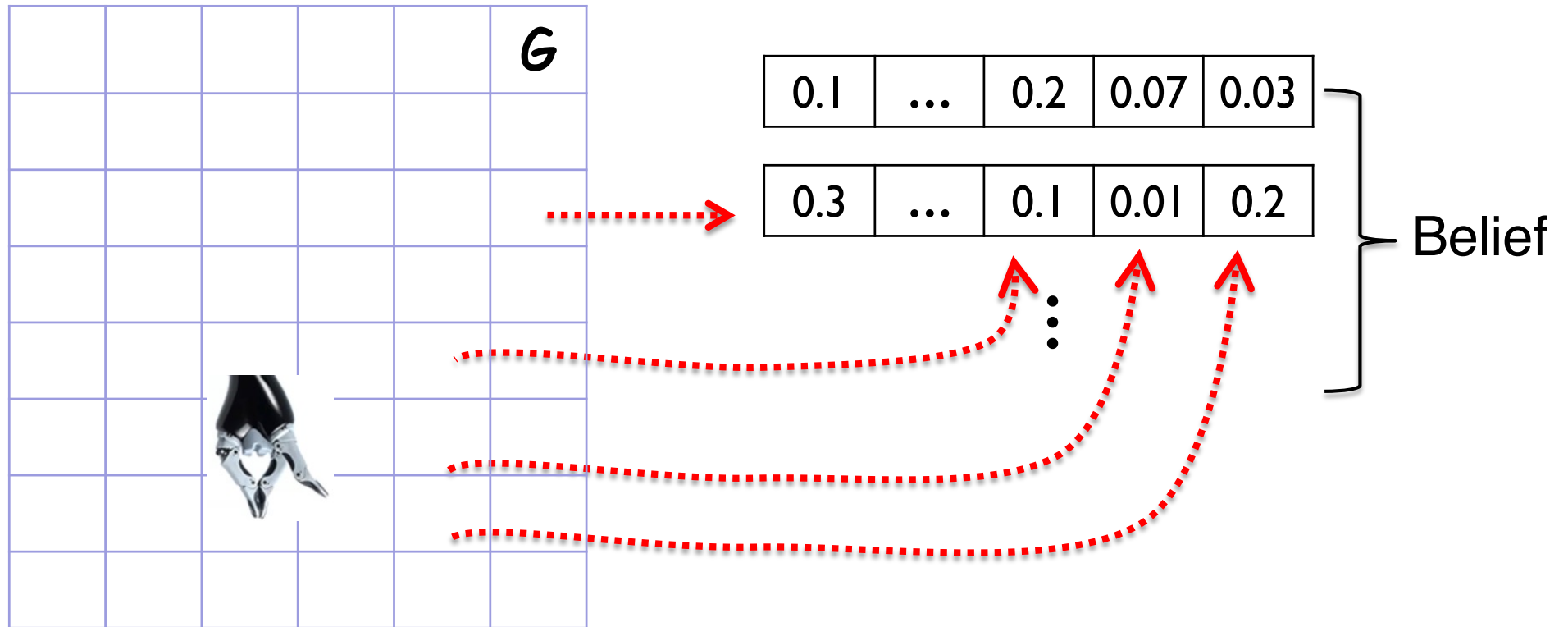
$$s, s' \in S, a \in A, o \in O$$

Key to POMDPs: Belief



- A belief: a distribution over the state space
- Belief space: The set of all beliefs.
 - What's the dimension of the belief space?

Key to POMDPs: Belief



- Belief: distribution over the state space
 - The belief space is a simplex with $\text{dim} = |S| - 1$
- Policy: mapping from beliefs to actions

Key to POMDPs: Belief

- A belief is a sufficient statistics of the entire history of actions performed and observations perceived
 - Hence, the 1st order Markov in POMDP is only in its model (transition and observation function)
 - A POMDP policy is computed wrt to the entire history because it is computed wrt beliefs
-

Best policy

- Maps each belief to an action that maximizes the following objective function

$$V^*(b) = \max_{a \in A} \left(\underbrace{\sum_{s \in S} R(s, a) b(s)}_{\text{Expected immediate reward}} + \gamma \underbrace{\sum_{o \in O} P(o|b, a) V^*(b')}_{\text{Expected total future reward}} \right)$$

b' : next belief after the system at belief b performs action a and observes o

γ : discount factor (0,1)

Computationally intractable [Papadimitriou & Tsikilis'87, Madani, et.al.'99].

Relation between MDPs & POMDPs

- A POMDP can be viewed as an MDP in the belief space

$$V^*(b) = \max_{a \in A} \left(\sum_{s \in S} R(s, a) b(s) + \gamma \sum_{o \in O} P(o|b, a) V^*(b') \right)$$

Immediate reward in the
belief space: $R(b, a)$

Total future reward in the
belief space:
 $\sum_{o \in O} P(b' | b, a, o) V(b')$

- MDP can be viewed as as a degenerate POMDP
-

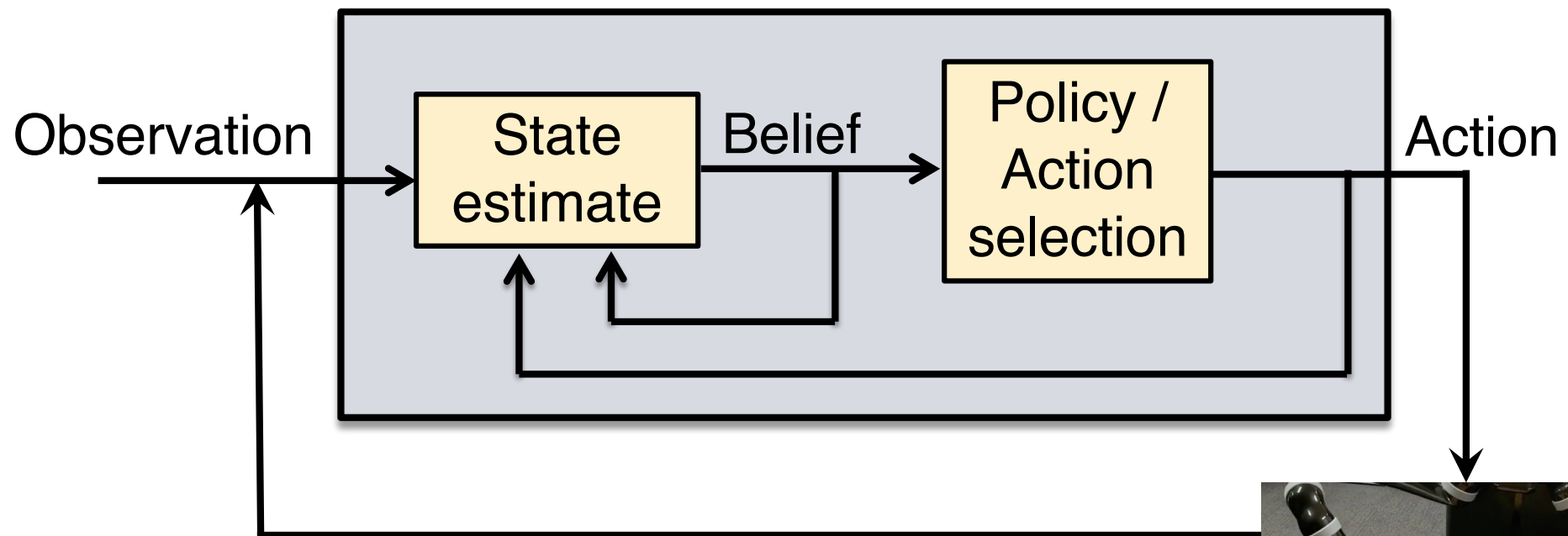
Formulation for b' and $P(o|b, a)$?

- Mathematically,
 - Use Bayes rule

$$\begin{aligned} b'(s') &= P(s'|o, a, b) \\ &= \frac{P(o|s', a, b)P(s'|a, b)\cancel{P(a, b)}}{P(o|a, b)\cancel{P(a, b)}} \\ &= \frac{P(o|s', a) \sum_s P(s'|s, a)b(s)}{\sum_{s''} P(o|s'', a) \sum_s P(s''|s, a)b(s)} \end{aligned}$$

The denominator $P(o|a, b)$ is essentially the normalizing factor that makes b' sums to 1.

Partially Observable Markov Decision Processes (POMDPs)



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Intuitively,

- Reinforcement learning (RL) is a machine learning approach that learns by doing
 - The agent is not provided with a model of how the robot's world works and/or which states are good
 - Rather, the agent learns by trial & error, and evaluation of states, actions, and their outcomes
-

Formally,

- An RL is an MDP where the transition and/or reward functions are not initially known

- S: State space.

- A: Action space.

- T: transition function.

- $$T(s, a, s') = P(S_{t+1} = s' \mid S_t = s, A_t = a).$$

- R: Reward function.

- $$R(s, a).$$

- Need to try & explore



At least one of these is not known

Solving?

- Two broad approaches
 - Model free: Learn the policy / value function directly
 - Model-based: Learn the MDP model, and then solve the MDP
-

RL can be framed as a POMDP

aka. Bayesian Reinforcement Learning

RL: MDP with missing components

- State space (S_{MDP})
- Action space (A_{MDP})
- Transition function (T_{MDP})
- Reward function (R_{MDP})

Not known

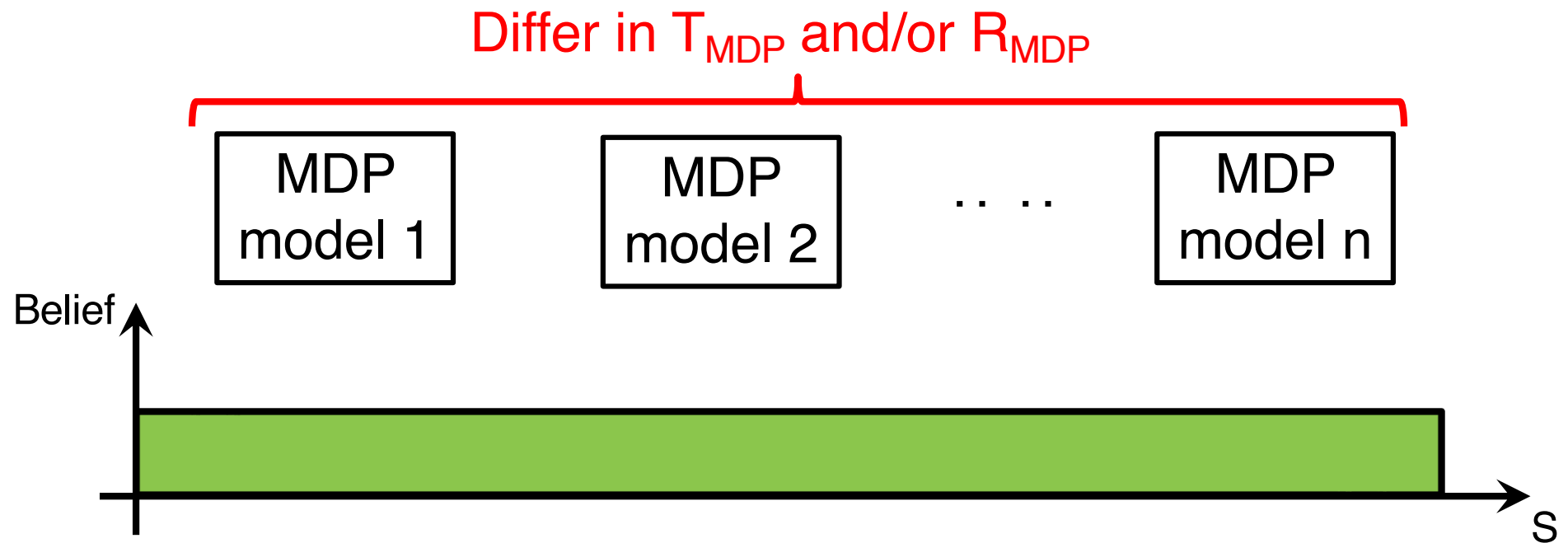
Bayesian RL

Construct a POMDP where

- The states are MDP states X parameters of the T_{MDP} & R_{MDP}
- Essentially, partial observability on which MDP model is the right model
- A, T, R follows from the particular MDP model
- O & Z are observations & observation function about which MDP model is correct

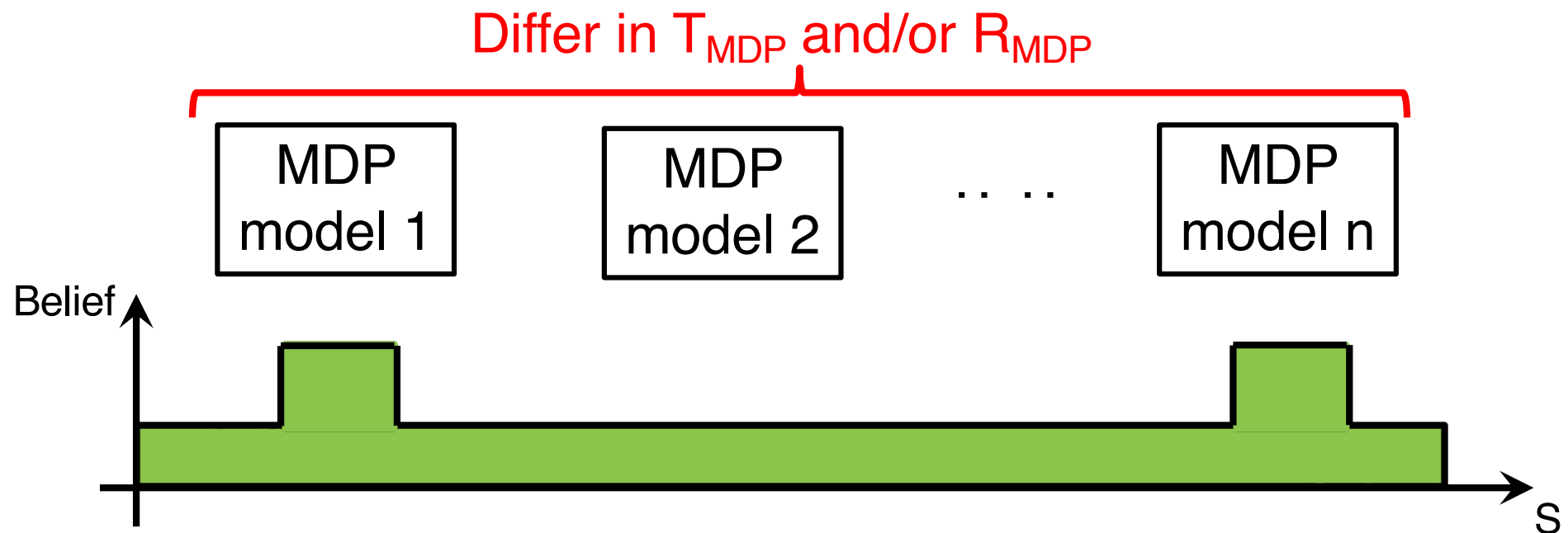
RL can be framed as a POMDP aka. Bayesian Reinforcement Learning

- Beliefs in POMDP becomes distribution over the possible MDP models



RL can be framed as a POMDP aka. Bayesian Reinforcement Learning

- Belief update means updates in the agent's understanding on which model is correct



POMDP automatically balances the trade-off between generating accurate model and to achieve the goal

In many cases, can achieve the goal without knowing the exact model

In a nutshell...

- MDP:
 - Non-deterministic action effects + fully observable
 - A degenerate POMDP
 - POMDP:
 - Non-deterministic action effects + partially observable
 - MDP in the belief space
 - RL:
 - MDP without transition and/or reward functions
 - Problem-wise, can be viewed as a POMDP, where partial observability is caused by incomplete information about the underlying MDP problem.
-

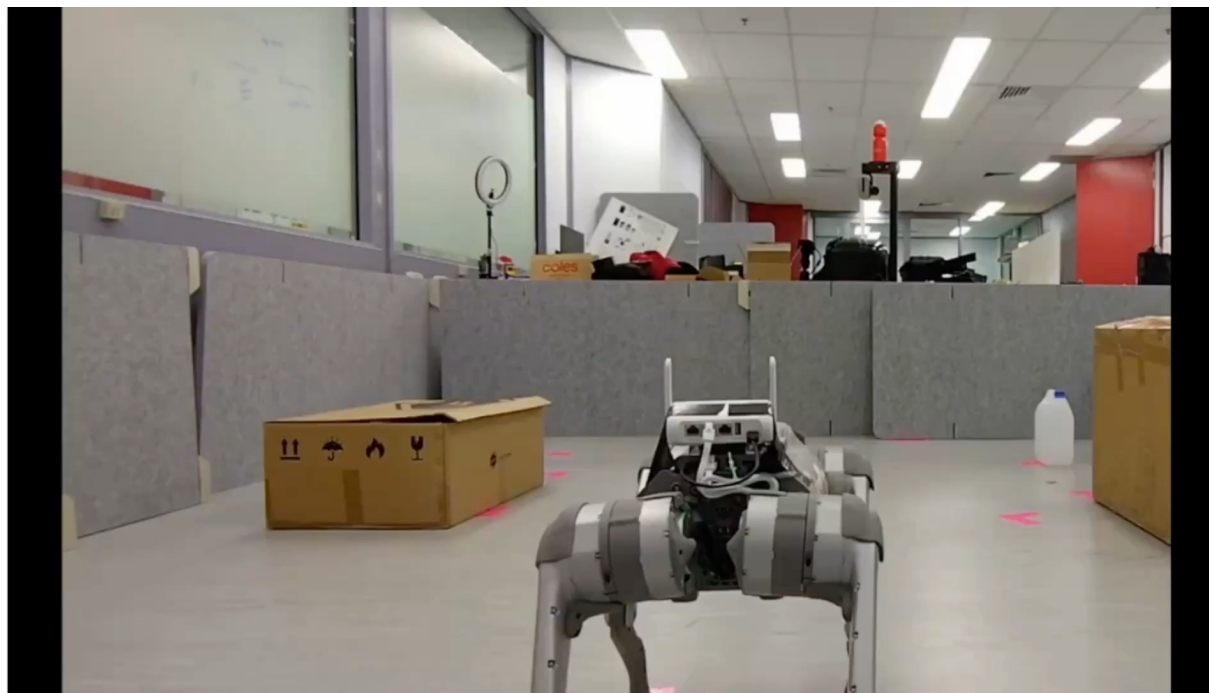
Why bother?

- Why not just give the data and learn the policy?
 - The structure helps learning
-

Model Learning + POMDP Planning on GPU

Target Finding in
Uncertain Environment

3x speed-up



Learn from simulation only, run direct on physical robot 10/10

Compare to Recurrent SAC (SOTA model free), use $< 10\%$ data for training, equal performance on seen environment, up to 7x higher success rate on unseen environments

M. Hoerger, M.R. Sudrajat, and H. Kurniawati. Vectorized Online POMDP Planning. ICRA'26

M. Hoerger, R. Joshi, R. Shome, I. Manchester, and H. Kurniawati. VOiLA: Vectorized Online planning with Learned diffusion model for POMDP Agents. Under review.

We finish the lecture here 😊

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Offline Solving

- Compute the entire policy before execution
- During execution, only need use the policy
 - Can think of the policy as a lookup table
 - Though, often using a more efficient representation than lookup table

Value Iteration for MDP

- Iterate calculating the optimal value of a state until convergence.

- Algorithm:

Initialize $V^0(s) = R(s, a)$ for all s

Loop

For all s {

$$V^{t+1}(s) = \max_a \left(R(s, a) + \gamma \sum_{s'} T(s, a, s') V^t(s') \right)$$

}

$t = t + 1$

Until $V^{t+1}(s) = V^t(s)$ for all s (impl: $\max_s |V^{t+1}(s) - V^t(s)| < 1e-7$)

- Will it converge to V^* ? Can we have a bound on how long / how many iterations to converge?

Often called value update or Bellman update or Bellman backup.



Value Iteration – Approximately Optimal

- No guarantee we'll reach optimal in finite time.
- However, it's guaranteed we can get close to optimal within finite time. In fact, polynomial in $|S|$, $|A|$, $1/(1-\gamma)$.
- If we want to ensure that we are ε close to optimal, then #iterations should be

$$\left\lceil \frac{\log(2R_{max}/\varepsilon(1-\gamma))}{\log(1/\gamma)} \right\rceil$$

Value Iteration – Issues

- Issues when the state space is large?
 - Computing Bellman update at every steps
 - Policy representation

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Online Solving

- Online Solving: At every step, the agent is given a short amount of time to compute the best action for the current step
 - Less memory requirements
 - A bit slower during execution, though these days, we can compute even good POMDP policy with 1 sec planning time per step for 6-DOFs manipulator
 - Does not need explicit model
 - Use a generative model (sampler, can be a simulator)
 - Computes a good policy by interacting with the simulator
 - Assume: Initial state/belief is known
-

Monte Carlo Tree Search (MCTS)

- Combines tree search & Monte Carlo
- Monte Carlo
 - A sampling-based method to approximate the value of complicated functions, i.e., the ones that are too difficult/time consuming to compute exactly.
 - In MDP, can be used to approximate the expected total future reward if the agent follows an optimal policy.

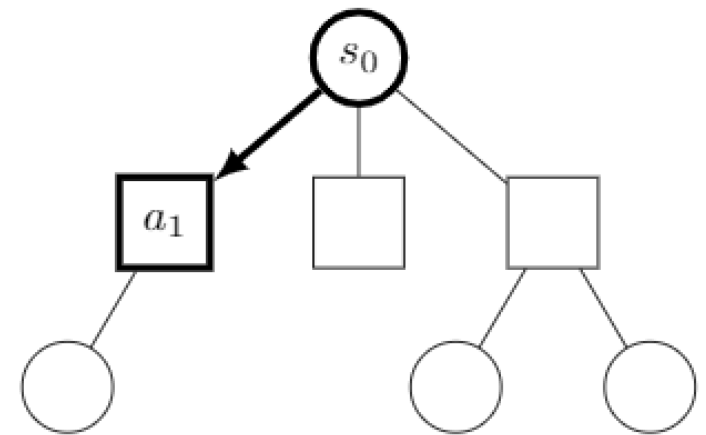
$$V_{\pi^*}(s) = R(s, a) + \gamma \sum_{s'} T(s, \pi(s), s') V_{\pi^*}(s')$$

Monte Carlo Tree Search (MCTS)

- AND-OR Tree
 - OR-nodes represent states & associated value estimates
 - AND-nodes represent actions & associated Q-value estimates
- Wait, why do we need AND-OR Tree for MDP, which is fully observed?

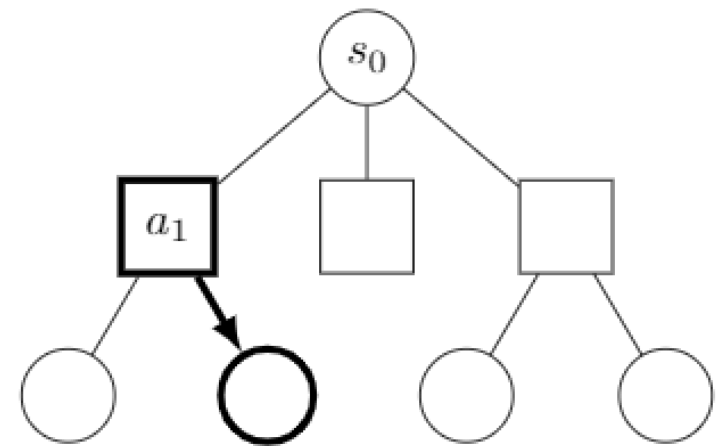
Monte Carlo Tree Search (MCTS)

- Build the search tree based on the outcomes of forward simulations
- Iterate over 4 main components:
 - Selection: Choose the best path



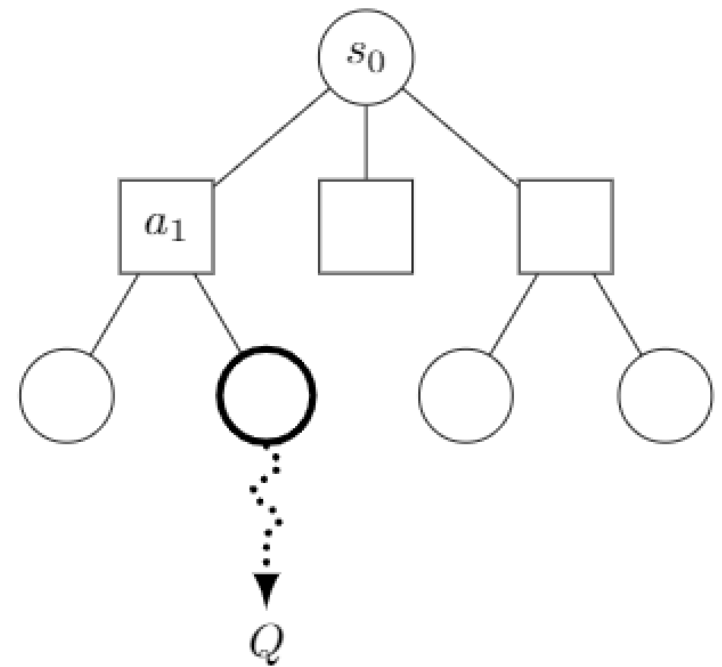
Monte Carlo Tree Search (MCTS)

- Build the search tree based on the outcomes of forward simulations
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 - Expansion: When a terminal node is reached, add a child node



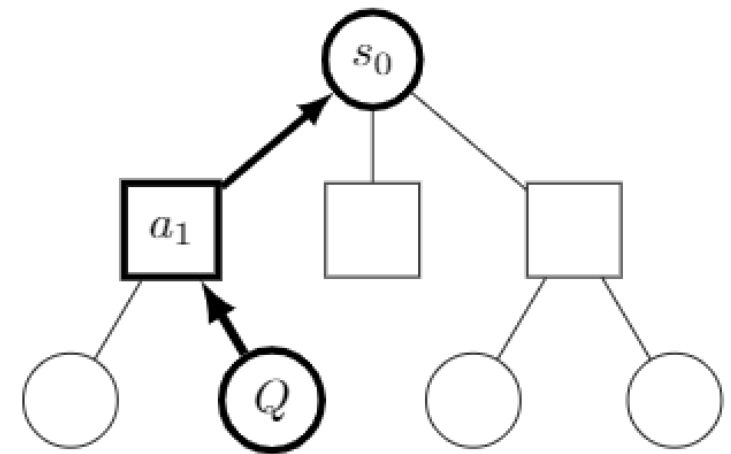
Monte Carlo Tree Search (MCTS)

- Build the search tree based on the outcomes of forward simulations
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 - Rollout: Estimate the value of the newly added node, usually via simulation but can also use learned functions



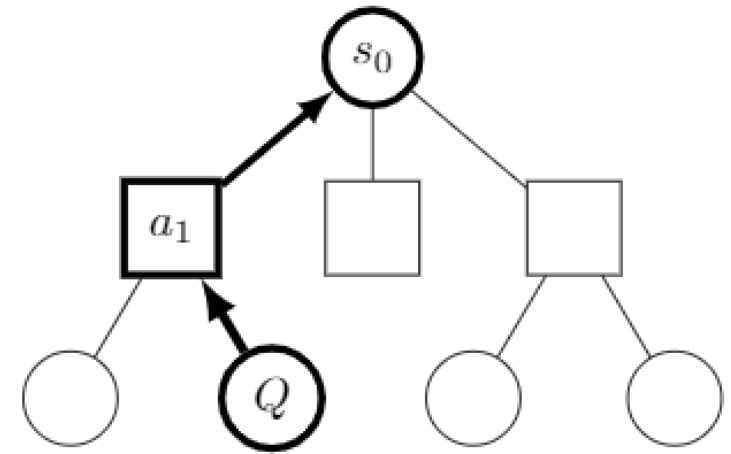
Monte Carlo Tree Search (MCTS)

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 - Backpropagation: Update the value of the nodes visited in this iteration



Commonly Used MCTS for MDP

- AND-OR Tree
- Build the search tree based on the outcomes of forward simulations
- Iterate over 4 main components:
 - Selection: Choose the best path
 - Expansion: When a terminal node is reached, add a child node
 - Rollout: Estimate the value of the newly added node, usually via simulation but can also use learned functions
 - Backpropagation: Update the value of the nodes visited in this iteration



Node Selection

- Multi-armed bandit to select which action to use
 - In general, use a method called Upper Confidence Bound (UCB1).

- Choose an action a to perform at s as:

$$\pi_{UCT}(s) = \arg \max_{a \in A} \underbrace{Q(s,a)}_{\text{Exploitation}} + c \underbrace{\sqrt{\frac{\ln(n(s))}{n(s,a)}}}_{\text{Exploration}}$$

C : A constant indicating how to balance exploration & exploitation, need to be decided by trial & error.

$n(s)$: #times node s has been visited.

$n(s,a)$: #times the out-edge of s with label a has been visited.

- MCTS + UCB is often called Upper Confidence Bound for Trees (UCT)

For approximate POMDP solving

H. Kurniawati. Partially Observable Markov Decision Processes and Robotics. *Annual Review of Control, Robotics, and Autonomous Systems*. Vol. 5. No. 1. 2022.

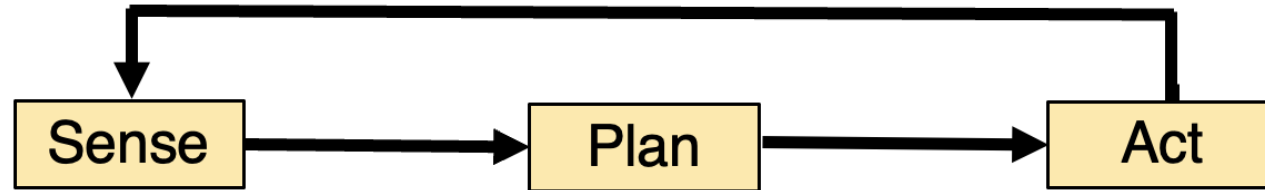
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-

What's Next?

- Today hands-on:
 - Motion Planning
 - Tomorrow workshop:
 - Motion planning with kinodynamic constraints
 - RL for low-level control
 - Integrating motion planning & low-level control
-

Take Home Message



When planning planning accounts for uncertainty, we can relax requirements for accuracy in model, which in turn reduce data & time to learn the model

Many types of uncertainty, but for planning, accounting for uncertainty in the effects of actions and in the observations are sufficient to cover the various types uncertainty

Of course, if we have more accurate model, we should be able to do more...

Exercises in Framing MDP/POMDP Problems

MDP: Simple Navigation

- An agent moves in 4X3 grid cells.
- It can move to one of four neighboring cells. The actions' accuracy is 70%.

			+1
			-1
S			

30% of the time, the agent ends up at the left or right of its intended cell, or at the current cell, with equal probability. If there's no cell in the left or right of its intended cell, the probability mass is added to staying where it is.

- Collision with obstacle/boundary will result in no movement.
- Two terminal states, with reward +1 and -1. All other actions incur a cost of -0.04.
- Please define the MDP framing of the above problem.

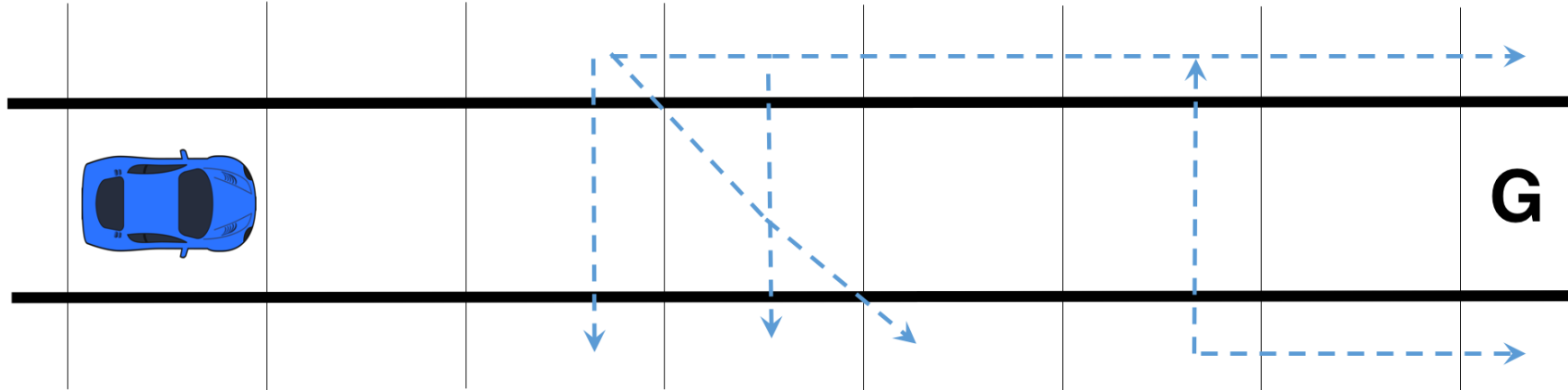
POMDP: Saving Luke Skywalker / Tiger

- Imagine R2D2 standing in a front of two closed doors. Behind one door is a tiger and behind the other is Luke Skywalker.
 - If R2D2 opens the door with the tiger, it will be injured and receive a penalty of -100. If it opens the door with Luke Skywalker behind the door, then R2D2 frees Luke and receives a large reward, i.e., +10.
 - Instead of opening the door directly, R2D2 can also gain information about the possible location (with some error) of the tiger by performing a LISTEN action. This action incurs a small cost of -1. Suppose the accuracy of the LISTEN action is 85%.
 - Suppose at the beginning (i.e., step-0), the tiger & Luke can be in any one of the rooms with equal probability.
 - The problem of deciding whether to listen or open the left or the right door so as to maximize the expected total reward, can be framed as a POMDP problem. Please define this POMDP framing.
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POMDP: Drones Navigation

- Consider the problem of centralised collision-free navigation of one or more drones in the presence of non-deterministic action effects, such as in the presence of wind disturbances
 - The drone operates underground and therefore does not receive GPS signals. However, It can localise well in certain regions of the environment and receives observation about the distance to its nearest obstacle with some error. This limited localisation capability makes the problem partially observed
 - How should we define a POMDP framing for this problem?
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POMDP: Pedestrian Avoidance



- Suppose a self-driving car has good enough sensors to identify its own position and the pedestrians' positions
 - However, pedestrians' intentions are not known before hand
 - How should the car behave to reach its goal without colliding with the pedestrians?
- .

Thank you
Q&A
